

$$A = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}, \quad X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$(A^T)^T = A$$

$$A^T X = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}^T \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = (a_1 \ a_2) \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = a_1 x_1 + a_2 x_2$$

$$R_p = w_1 R_1 + w_2 R_2$$

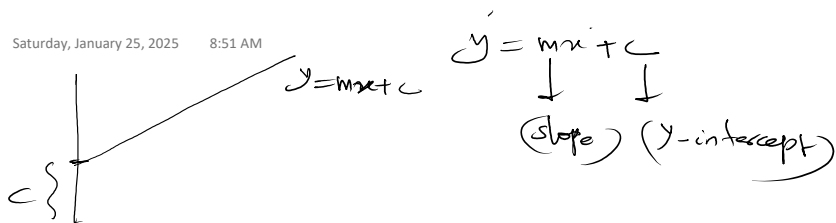
$$W = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}$$

$$R = \begin{pmatrix} R_1 \\ R_2 \end{pmatrix}$$

$$R_p = W^T R$$

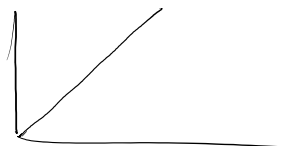
$$V(A^T X) = A \Omega A^T$$

$$V(R_p) = W^T \Omega W$$



$$\underline{x} = a(\underline{t}) \quad \frac{y}{x} = \left(\frac{b}{a}\right)$$

$$\underline{y} = b(\underline{t}) \quad y = \left(\frac{b}{a}\right)x$$



$$\underline{\mu}_p = \underline{r}_f + (\underline{w}_A)(\underline{\mu}_A - \underline{r}_f)$$

$$\underline{\sigma}_p = (\underline{w}_A)\underline{\sigma}_A$$

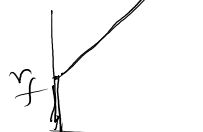
slope

$$\underline{\mu}_p = \underline{r}_f + \frac{\underline{\sigma}_p}{\underline{\sigma}_A}(\underline{\mu}_A - \underline{r}_f)$$

$$\underline{\mu}_p = \text{Const.} + \text{slope} \times \underline{\sigma}_p$$

$\downarrow$ $\downarrow$

$\underline{r}_f$ $\left(\frac{\underline{\mu}_A - \underline{r}_f}{\underline{\sigma}_A}\right)$



$$\sigma_p^2 = \underbrace{w_1^2 \sigma_1^2} + \underbrace{w_2^2 \sigma_2^2} + 2w_1 w_2 \rho \sigma_1 \sigma_2$$

Case I. $\rho = 1$

$$\sigma_p^2 = (w_1 \sigma_1 + w_2 \sigma_2)^2$$

$$\sigma_p = |w_1 \sigma_1 + w_2 \sigma_2| \quad \checkmark$$

Case II - $\rho = -1$

$$\sigma_p^2 = (w_1 \sigma_1 - w_2 \sigma_2)^2$$

$$\sigma_p = |w_1 \sigma_1 - w_2 \sigma_2| \quad \checkmark$$

$$\sqrt{x^2} = |x|$$

$$\left. \begin{array}{l} \sqrt{5^2} = 5 \\ \sqrt{(-5)^2} = 5 \end{array} \right\}$$

$$|x|$$

$$\sigma_P^2 = w_A^2 \sigma_A^2 + (1-w_A)^2 \sigma_B^2 + 2w_A(1-w_A)\rho\sigma_A\sigma_B$$

$$2\sigma_P \cdot \left(\frac{d\sigma_P}{dw_A} \right) \overset{0}{=} 2w_A \cdot \underline{\sigma_A^2} - 2(1-w_A) \underline{\sigma_B^2} + 2\rho \underline{\sigma_A} \underline{\sigma_B} (1-2w_A)$$

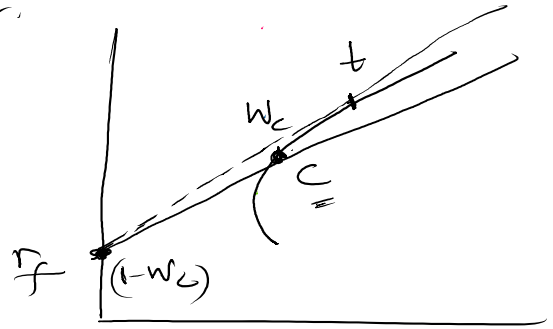
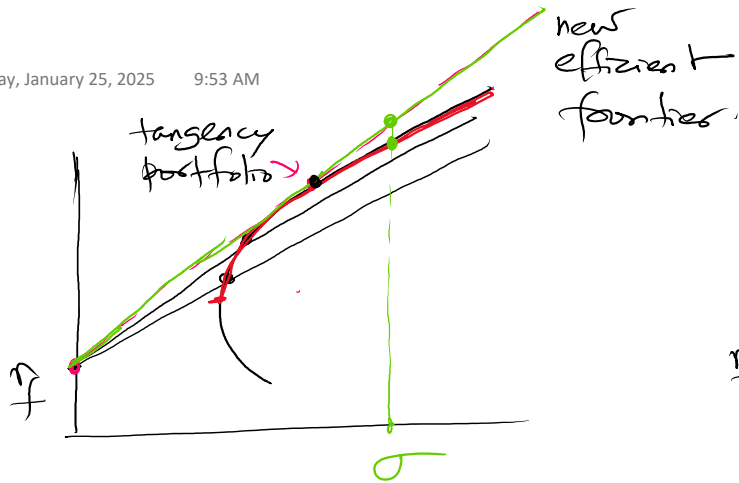
$$\Rightarrow w_A (\sigma_A^2 + \sigma_B^2 - 2\rho\sigma_A\sigma_B) - \sigma_B^2 + \rho\sigma_A\sigma_B = 0$$

$$\Rightarrow \underline{w_A} = \frac{\sigma_B(\sigma_B - \rho\sigma_A)}{\sigma_A^2 + \sigma_B^2 - 2\rho\sigma_A\sigma_B}$$

Case I, $\rho = 1$ (no diversification)

$$\mu_p =$$

$$\sigma_p =$$



$$\begin{aligned}\mu_p &= (1-w_C)r_f + w_C\mu_C \\ &= r_f + (w_C)(\mu_C - r_f)\end{aligned}$$

$$\sigma_p = (w_C) \times \sigma_C$$

$$\mu_p = r_f + \left(\frac{\mu_C - r_f}{\sigma_C} \right) \cdot \sigma_p$$

$$\mu_p = r_f + S_C \times \sigma_p$$

↑
slope = Sharpe ratio of C

tangency portfolio is the one with highest Sharpe ratio

$$S_+ = \left(\frac{\mu_+ - r_f}{\sigma_+} \right)$$